

Analysis First-Year Exam, January 2026, Part 1

Answer FOUR questions in detail. State carefully any results used without proof.

1. Prove

- (a) that every real sequence has a monotonic subsequence
- (b) that every bounded real sequence has a convergent subsequence.

2. Let T be a two-point metric space. Prove that the metric space X is connected if and only if each continuous function from X to T is constant.

3. Let $X \subseteq \mathbb{R}$ and consider the following statements: Does (a) imply (b)? Does (b) imply (a)? Proof or counterexample for each.

- (a) each continuous function from X to $\mathbb{R}$ is bounded;
- (b) X is compact.

4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be uniformly continuous and let $B \subseteq \mathbb{R}$ be bounded. Must $f(B)$ be bounded? Proof or counterexample.

5. The continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable except perhaps at the point a . Decide whether f is necessarily differentiable at a if it is further assumed that $f'(t) \rightarrow 0$ as $t \rightarrow a$.